
Taylor-Made Sparsity: First-Order Approximation for Structural Pruning in Object Detectors

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Abstract

Deep neural networks have achieved strong performance in object detection, but their computational cost limits deployment on resource-constrained edge devices. Structured pruning offers a practical solution by removing entire channels or filters to reduce FLOPs and latency. However, commonly used magnitude-based criteria (e.g., L1-norm) are data-independent and often fail to reflect the true functional contribution of network components, leading to significant accuracy degradation at higher sparsity levels. On the other hand, more principled attribution methods based on second-order optimization or game-theoretic formulations are computationally prohibitive for modern detection architectures.

We propose **Taylor-Made Sparsity**, a fast and data-driven structured pruning framework based on a first-order Taylor approximation of the loss function. Our method estimates channel importance using the squared gradient–activation interaction, $(\nabla L \odot A)^2$, accumulated over a small calibration subset. This provides an efficient proxy for functional sensitivity that computes global importance scores across all layers of a deep network in approximately 1.5 seconds using standard backpropagation.

To ensure deployable efficiency, we integrate this sensitivity criterion with group-aware structural pruning, enabling consistent tensor reshaping and preserving hardware alignment. This ensures that theoretical reductions in parameters and FLOPs translate into practical inference speedups.

Extensive experiments on ResNet-50 (CIFAR-10) and MobileNetV2-SSD (MS COCO 2017) demonstrate that the proposed approach consistently improves the accuracy–efficiency trade-off over magnitude-based baselines. Our method remains stable at pruning ratios up to 50%, avoiding the severe degradation observed with conventional heuristics while maintaining high inference throughput.

1 Introduction

The deployment of deep neural networks (DNNs) for object detection has driven transformative advances in autonomous navigation, smart surveillance, and mobile vision. However, the architectural complexity required to achieve high mean Average Precision (mAP) imposes severe computational and memory constraints. Deploying these models on resource-restricted edge devices, such as microcontrollers or unmanned aerial vehicles (UAVs), requires strict adherence to tight floating-point operation (FLOP) and memory budgets [Tran et al., 2026]. While lightweight architectures like MobileNetV2 [Sandler et al., 2018], Single Shot Detectors (SSD) [Liu et al., 2016], and modern attention-centric models like YOLOv12 [Tian et al., 2025] offer efficient baselines, they still harbor task-specific redundancy that necessitates post-training compression.

Structured pruning has emerged as a practical solution for edge deployment because it physically removes entire channels or filters, translating directly into inference speedups on commodity hardware without requiring specialized sparse-tensor accelerators [Fang et al., 2023]. The central challenge in structured pruning is determining the importance criterion—*what* to prune. Traditional methods heavily rely on magnitude-based heuristics, such as the L1-norm of the weights, assuming that small weights are functionally negligible [Li et al., 2017]. While computationally cheap, this data-independent heuristic frequently triggers catastrophic network collapse at extreme sparsity levels (e.g., $> 30\%$) by indiscriminately removing structurally small but functionally critical bottleneck layers [Shukla and Tapas, 2026].

To overcome this, recent literature has explored sensitivity-aware criteria, leveraging Explainable AI (XAI) and Taylor expansions to estimate a parameter’s contribution to the loss function [Molchanov et al., 2017]. However, scaling these approaches to modern object detectors introduces severe bottlenecks. Computing exact second-order Hessian information (e.g., Optimal Brain Damage [LeCun et al., 1989, Hassibi and Stork, 1993]) or cooperative game-theoretic attributions (e.g., exact SHAP values) is mathematically elegant but computationally intractable for layer-wise assessment in deep, dense vision pipelines. Furthermore, even when an accurate importance criterion is found, applying naive layer-wise pruning to lightweight detectors frequently shatters depthwise convolution groups, creating memory misalignments that severely bottleneck hardware inference speeds. Unlike prior Taylor-based pruning methods that operate in a layer-wise or iterative manner, our approach performs a globally calibrated importance estimation across all layers in a single pass, enabling practical deployment under strict latency and compute constraints.

In this work, we introduce **Taylor-Made Sparsity**, a fast, data-driven structured pruning framework that optimizes the accuracy–efficiency Pareto frontier without requiring retraining or architecture redesign. By formalizing pruning as a loss-preservation problem, we leverage a proxy inspired by the first-order Taylor approximation—specifically, a squared gradient–activation interaction $(\nabla L \odot A)^2$ —to quantify the functional sensitivity of feature maps. To ensure these theoretical FLOP reductions translate into real-world inference speedups, we seamlessly integrate our Taylor criteria with group-aware dependency modeling, guaranteeing hardware alignment. While we report throughput on a high-end GPU for controlled benchmarking, the observed trends reflect structural efficiency improvements rather than device-specific optimizations.

Our main contributions are summarized as follows:

- **Highly Efficient First-Order Approximation:** We propose a squared gradient–activation proxy that effectively captures a reliable estimate of the functional sensitivity of convolutional channels. Utilizing PyTorch forward hooks over a strictly isolated calibration subset, this data-driven proxy computes global importance scores in just ~ 1.5 seconds, bypassing the intractability of second-order derivatives.
- **Hardware-Aware Structural Constraints:** We integrate our sensitivity criterion with automated dependency graph modeling. This enforces group-aware pruning that strictly preserves depthwise tensor alignments, successfully avoiding the unoptimized memory-access crashes that plague naive structural pruning on edge hardware.
- **Robustness at Extreme Sparsities:** Through comprehensive experiments on ResNet-50 (CIFAR-10) and MobileNetV2-SSD (MS COCO 2017), we demonstrate that our framework establishes a strong accuracy–efficiency Pareto frontier. Our method scales gracefully up to 50% sparsity, consistently outperforming magnitude-based baselines in feature retention while maintaining high inference throughput (up to ~ 150 FPS under hardware-aware constraints).

Code and pretrained models to reproduce all experiments are provided in the supplementary material.

2 Related Work

Efficient Object Detection for Edge Deployment. Object detection has evolved from proposal-based two-stage frameworks such as Faster R-CNN [Ren et al., 2015] to dense single-stage detectors such as SSD [Liu et al., 2016], which reduce latency through unified localization and classification. Subsequent YOLO variants further improved the speed–accuracy trade-off for real-time applications,

88 while recent attention-integrated designs (e.g., YOLOv12) demonstrate that transformer-style compo-
89 nents can be incorporated without sacrificing efficiency [Tian et al., 2025]. Lightweight backbones
90 such as MobileNetV2 and ShuffleNet remain standard choices for embedded deployment due to
91 their favorable compute characteristics [Sandler et al., 2018, Zhang et al., 2018]. Nevertheless, even
92 compact detectors exhibit task-specific redundancy after training, motivating post-hoc compression
93 techniques.

94 **Detector Compression under Resource Constraints.** Deploying detection models on microcon-
95 trollers and low-power devices has driven research into hardware-aware design and compression.
96 Approaches such as TinyissimoYOLO emphasize ultra-compact architectures through aggressive
97 quantization [Moosmann et al., 2023], while ELASTIC formulates once-for-all architecture search
98 tailored to microcontroller constraints [Tran et al., 2026]. Although these approaches optimize archi-
99 tecture design, they often require expensive search or retraining pipelines. In contrast, pruning-based
100 methods provide a complementary pathway by adapting pretrained models to specific deployment
101 budgets with minimal overhead.

102 **Structural Pruning and Dependency Modeling.** Structured pruning removes entire channels or
103 filters, enabling direct reductions in FLOPs and latency on standard hardware. Early approaches
104 relied on magnitude-based criteria (e.g., L1 norms) [Li et al., 2017], which are computationally
105 efficient but ignore task-dependent sensitivity. Modern architectures introduce complex dependencies
106 through residual connections, concatenation, and multi-scale fusion, making naive pruning unsafe.
107 Dependency-graph-based methods, such as DepGraph [Fang et al., 2023], address this by explicitly
108 modeling structural couplings and enabling safe pruning across arbitrary network topologies. Subse-
109 quent frameworks extend this idea to broader architectural settings [Wang et al., 2024, Sundaram
110 et al., 2025]. However, these methods primarily focus on *how* to prune safely, leaving the choice of
111 importance criterion largely orthogonal.

112 **Taylor and Sensitivity-Based Pruning.** A principled pruning objective is to remove components
113 whose removal minimally increases the loss. Classical second-order approaches (e.g., Optimal Brain
114 Damage and Optimal Brain Surgeon) approximate this effect using Hessian information but are
115 computationally prohibitive for modern deep networks [LeCun et al., 1989, Hassibi and Stork, 1993].
116 First-order approximations provide a scalable alternative. In particular, Molchanov et al. showed
117 that gradient-activation interactions offer an efficient proxy for estimating the loss increase caused
118 by removing convolutional filters [Molchanov et al., 2017]. While these methods improve over
119 magnitude-based heuristics, most prior work focuses on image classification and does not directly
120 address the structural and multi-branch complexities of object detection pipelines.

121 **Attribution-Guided Compression.** Explainability methods have also been explored for model
122 compression. Techniques such as DeepLIFT estimate feature relevance through activation differences
123 [Shrikumar et al., 2017], and subsequent work has leveraged attribution signals for pruning and model
124 simplification. Recent approaches apply explainability-driven criteria to object detection, demonstrat-
125 ing improved robustness over magnitude-based pruning [Shukla and Tapas, 2026]. However, these
126 methods typically operate in a layer-wise manner and do not integrate structural constraints required
127 for hardware-aligned deployment.

128 **Position of This Work.** Our approach bridges sensitivity-based pruning and deployment-aware
129 structural pruning. Unlike architecture-search-based methods [Tran et al., 2026], we operate directly
130 on pretrained detectors without retraining or search. Compared with dependency-aware frameworks
131 [Fang et al., 2023], we introduce a data-driven importance criterion tailored to dense detection.
132 Finally, in contrast to prior Taylor-based pruning [Molchanov et al., 2017], which is predominantly
133 classification-oriented or layer-wise, we perform a globally calibrated importance estimation across
134 all layers in a single pass and integrate it with dependency-aware constraints. This combination yields
135 a practical and efficient framework for improving the accuracy-efficiency trade-off of modern object
136 detectors under real deployment constraints.

3 Methodology

To enable efficient compression of object detection architectures without structural collapse, a pruning framework must address two key questions: *what* to prune (importance estimation) and *how* to prune (structural consistency). We propose a first-order Taylor-based importance proxy combined with dependency-aware structural constraints to ensure both accuracy preservation and deployable efficiency.

3.1 Problem Formulation and Taylor Approximation

We formulate structured pruning as minimizing the change in the loss function $\mathcal{L}$ when removing a feature map. Let $A_i^{(l)} \in R^{H \times W}$ denote the i -th activation map of layer l . Removing this feature map corresponds to setting it to zero, yielding a perturbation:

$$\delta A_i^{(l)} = -A_i^{(l)}. \quad (1)$$

Following prior work [LeCun et al., 1989], the change in loss can be approximated via a first-order Taylor expansion:

$$\Delta \mathcal{L}(A_i^{(l)}) \approx \left\langle \frac{\partial \mathcal{L}}{\partial A_i^{(l)}}, \delta A_i^{(l)} \right\rangle + \mathcal{O}(\|\delta A\|^2), \quad (2)$$

where $\langle \cdot, \cdot \rangle$ denotes the Frobenius inner product over spatial dimensions.

Substituting $\delta A_i^{(l)} = -A_i^{(l)}$ and taking the absolute value:

$$|\Delta \mathcal{L}(A_i^{(l)})| \approx \left| \sum_{h,w} \frac{\partial \mathcal{L}}{\partial A_i^{(l)}(h,w)} \cdot A_i^{(l)}(h,w) \right|. \quad (3)$$

This yields a scalar importance estimate by explicitly aggregating over spatial dimensions, ensuring mathematical consistency without requiring vectorization.

3.2 Taylor-Made Importance Criterion

Single-sample gradient estimates are noisy, particularly in object detection where classification and localization losses interact. To obtain a stable importance estimate, we compute expectations over a calibration subset D_{cal} .

Instead of using the signed first-order estimate directly, we adopt a squared formulation to reduce gradient sign cancellation and emphasize consistently high-magnitude interactions. The importance score for the i -th channel in layer l is defined as:

$$S_{l,i}^{\text{Taylor}} = E_{\mathbf{x} \sim D_{\text{cal}}} \left[\sum_{h,w} \left(\nabla_{A_i^{(l)}} \mathcal{L}(\mathbf{x}) \odot A_i^{(l)}(\mathbf{x}) \right)^2 \right]. \quad (4)$$

Rationale. In dense object detection, directly aggregating first-order gradient interactions across spatial locations often leads to sign cancellation. Gradients from classification and localization objectives may exhibit opposing signs across neighboring pixels, particularly between foreground and background regions. This phenomenon can be explicitly illustrated as:

$$\sum_{h,w} \nabla_A \mathcal{L}(h,w) \odot A(h,w) \approx \sum_{h,w} (g^+(h,w) - g^-(h,w)) A(h,w) \approx 0, \quad (5)$$

where $g^+(h,w) = \max(\nabla_A \mathcal{L}(h,w), 0)$ and $g^-(h,w) = -\min(\nabla_A \mathcal{L}(h,w), 0)$ denote positive and negative gradient contributions. This illustrates how opposing gradient contributions across spatial locations can cancel, leading to underestimation of channel importance.

By squaring the interaction inside the spatial summation, our formulation eliminates this cancellation effect and instead accumulates the magnitude of pixel-wise sensitivities. This yields a more stable estimate of channel importance in multi-task settings.

170 **Connection to Second-Order Sensitivity.** The squared gradient-based formulation is also consistent
 171 with second-order sensitivity measures. In particular, the expected squared gradient is closely related
 172 to the diagonal of the empirical Fisher Information Matrix (FIM), which serves as a positive semi-
 173 definite approximation to the Hessian under standard assumptions [LeCun et al., 1989, Hassibi and
 174 Stork, 1993].

175 From this perspective, the proposed proxy can be interpreted as a computationally efficient, first-order
 176 approximation that captures second-order curvature information. While not an exact Hessian estimate,
 177 it provides a stable and practically effective signal for channel importance, especially in high-variance
 178 loss landscapes such as object detection.

179 Using standard forward and backward hooks, this score is computed across all layers in a single pass
 180 over D_{cal} , requiring approximately ~ 1.5 seconds for ResNet-50.

181 3.3 Dependency-Aware Structural Pruning

182 Modern detection architectures contain complex structural dependencies (e.g., residual connections,
 183 concatenation, and multi-scale fusion). Pruning individual channels independently may break tensor
 184 compatibility.

185 We adopt the dependency graph formulation [Fang et al., 2023], where the network is decomposed
 186 into components $F = \{f_1, \dots, f_L\}$. Dependencies between components are encoded as:

$$D(f_i^-, f_j^+) = 1[f_i^- \leftrightarrow f_j^+] \vee 1[i = j \wedge \text{sch}(f_i) = \text{sch}(f_j)], \quad (6)$$

187 where $\leftrightarrow$ denotes structural coupling (e.g., residual connections, concatenation, or shared channel
 188 dimensions), and $\text{sch}(\cdot)$ denotes the pruning scheme associated with a component (i.e., the rule
 189 governing how its channels are removed or retained).

190 The network is partitioned into groups $\mathcal{G}$ of mutually dependent channels. We aggregate importance
 191 scores within each group using summation:

$$S(\mathcal{G}_k) = \sum_{(l,i) \in \mathcal{G}_k} S_{l,i}^{\text{Taylor}}. \quad (7)$$

192 Groups with the lowest scores are pruned synchronously, ensuring structural consistency and valid
 193 tensor shapes.

194 3.4 Hardware-Aware Constraints

195 While dependency graphs ensure structural validity, they do not guarantee efficient hardware execution.
 196 We introduce two practical constraints:

197 **Depthwise Convolution Alignment.** Depthwise convolutions require channel counts divisible
 198 by the group size. We enforce a `round_to = 8` alignment and avoid directly pruning depthwise
 199 layers. Instead, they are adjusted indirectly through adjacent pointwise convolutions, preserving
 200 kernel efficiency and memory alignment.

201 **Detection Head Preservation.** Detection heads are tightly coupled with post-processing operations
 202 (e.g., Non-Maximum Suppression). Pruning these components may introduce memory misalignment
 203 and degrade inference speed. We therefore exclude detection heads from pruning and restrict
 204 modifications to the feature extraction backbone.

205 These constraints ensure that theoretical FLOP reductions translate into consistent real-world
 206 speedups.

207 4 Experiments and Results

208 To rigorously evaluate the proposed **Taylor-Made Sparsity** framework, we conduct extensive
 209 experiments across two distinct vision tasks: image classification (ResNet-50 on CIFAR-10) and
 210 multi-scale object detection (MobileNetV2-SSD on MS COCO 2017). Our evaluation isolates two
 211 primary dimensions of structural pruning: the empirical efficacy of the importance criterion (scoring)
 212 and the physical deployment viability on edge hardware (execution).

4.1 Experimental Setup

Baselines: We compare our Taylor-based Gradient-Activation proxy (Grad-Act) against the traditional L1-Norm (magnitude-based) heuristic [Li et al., 2017] to evaluate criteria efficacy under strict, constraint-free channel reduction. Additionally, we benchmark against the state-of-the-art Dependency Graph (DepGraph) framework [Fang et al., 2023] to evaluate structural enforcement. All methods are evaluated under identical pruning ratios and are followed by a standardized 5-epoch post-pruning fine-tuning phase. We adopt this short fine-tuning schedule to isolate the structural robustness of pruning methods rather than their full retraining capacity. This setting emphasizes immediate performance retention after pruning, providing a controlled comparison across methods. Longer fine-tuning schedules may further improve absolute performance but are orthogonal to our analysis. All results are averaged over 3 runs.

Metrics: Performance is measured via Top-1 Accuracy (classification) and mAP@[.50:.95] (detection). Efficiency is measured theoretically via GFLOPs and parameter counts, and practically via inference throughput (Frames Per Second, FPS). All throughput measurements are conducted on an NVIDIA RTX 4090 GPU for controlled benchmarking. While absolute latency may differ on edge devices, the observed trends reflect structural efficiency improvements rather than device-specific optimizations.

4.2 Evaluation of Importance Criteria

We first evaluate the pure efficacy of the importance criteria by applying strict, constraint-free pruning to the ResNet-50 architecture (baseline: 97.12% Acc, 1.31 GFLOPs).

As shown in Figure 1, at low sparsity levels (5–20%), the network retains sufficient capacity to absorb the loss of filters, resulting in similar performance across metrics. However, the true test of an importance criterion lies in the high-sparsity regime. At 30% nominal channel sparsity, our Grad-Act method achieves 94.84% Top-1 accuracy, outperforming the L1-Norm baseline (94.30%) by a +0.54% margin. This empirical gap widens at extreme compressions. The divergence of these curves confirms our hypothesis: static magnitude heuristics misjudge structurally small but functionally critical bottleneck layers, whereas our first-order Taylor approximation dynamically identifies and protects them, preventing catastrophic network collapse.

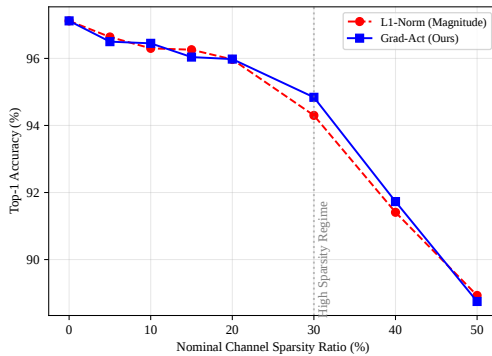


Figure 1: Accuracy vs. Sparsity on ResNet-50. At high sparsity regimes ($> 20\%$), the L1-Norm magnitude heuristic triggers rapid network degradation, while our Taylor-based proxy accurately preserves functionally critical features.

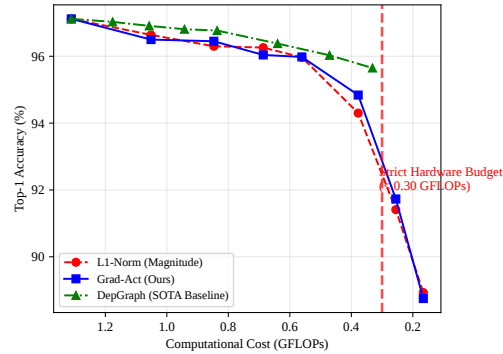


Figure 2: Pareto frontier of Accuracy vs. Computational Cost. Unlike dependency-graph methods that retain higher parameter counts to maintain accuracy, our method enforces strict structural sparsity, satisfying a strict < 0.30 GFLOP budget.

4.3 Comparison Under Strict Computational Constraints

While state-of-the-art structural frameworks like DepGraph exhibit the lowest absolute accuracy degradation at high nominal pruning ratios (e.g., 95.65% accuracy at 50% nominal sparsity), Figure 2 reveals that this comes at the direct expense of computational reduction.

While both methods utilize dependency graph constraints, our approach differs significantly in the pruning objective. DepGraph prioritizes structural safety, whereas our Taylor-based scoring identifies more aggressively redundant channels within valid groups, leading to lower effective FLOPs under the same pruning ratio.

If a deployed microcontroller imposes a strict hardware budget of < 0.30 GFLOPs (indicated by the red threshold in Figure 2), DepGraph fails entirely at the 50% ratio, requiring 0.33 GFLOPs and retaining 5.90M parameters. In stark contrast, our Grad-Act method satisfies the constraint at 0.26 GFLOPs at 40% and reaches 0.17 GFLOPs (3.22M parameters) at 50%. Our formulation provides a principled and empirically effective pathway that enables consistent FLOP reduction on highly constrained edge devices.

4.4 Object Detection on Edge: Hardware-Aware Execution

Compressing lightweight multi-scale object detectors (e.g., edge-scaled MobileNetV2-SSD) introduces severe architectural challenges, as the baseline model’s dense feature priors (0.210 mAP, 1.224 GFLOPs, ~ 148 FPS) are highly sensitive to layer-wise disruptions.

Efficacy of the Taylor Proxy: At low-to-medium sparsity regimes, our Grad-Act proxy demonstrates its superiority over magnitude-based heuristics on MS COCO. At a 5% pruning ratio, Grad-Act achieves an mAP of 0.1438, outperforming L1-Norm (0.1421) and the baseline DepGraph score (0.1429) under an identical computational constraint of 1.141 GFLOPs. This confirms that the squared gradient-activation product better captures task-specific spatial priors that static weights ignore.

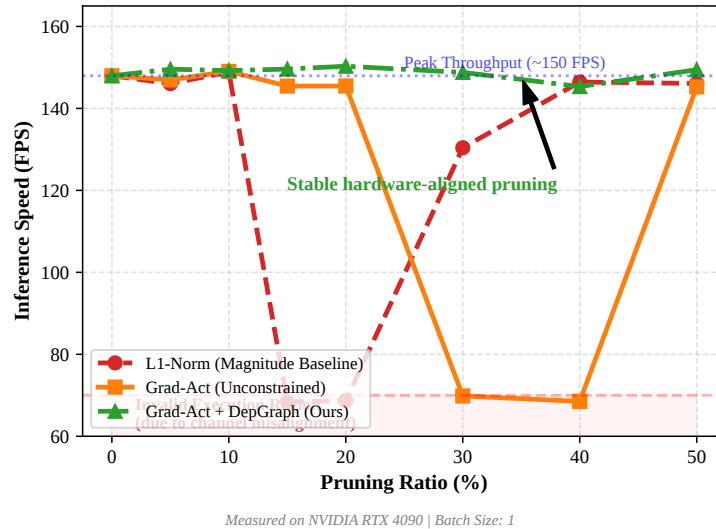


Figure 3: Inference Speed (FPS) vs. Pruning Ratio on MobileNetV2-SSD. Naive layer-wise pruning creates misaligned tensor dimensions that break CuDNN optimizations, resulting in a region of significant inference slowdown. Routing our Taylor scores through a group-aware dependency framework maintains hardware alignment and a stable ~ 150 FPS on an NVIDIA RTX 4090 GPU.

Inference Slowdown at High Sparsity: As sparsity increases, the physical execution of the pruning mask dominates deployability. As depicted in Figure 3, naive constraint-free pruning methods suffer from severe hardware slowdowns. Specifically, L1-Norm plunges into a region of significant inference slowdown (“Hardware Crash Zone”) of ~ 68.4 FPS at 15% sparsity, while constraint-free Grad-Act drops to 68.5 FPS at 40%. The curve labeled “Grad-Act (Unconstrained)” in Figure 3 corresponds to the constraint-free pruning variant. For these unconstrained pruning baselines (e.g., L1-Norm and Grad-Act without dependency modeling), we report runtime using zero-masked channels rather than physically pruned tensors. This avoids runtime errors due to tensor mismatches while reflecting the practical slowdown caused by unstructured memory access patterns.

Synthesis for Edge Deployment: These empirical failures validate the hardware-aware constraints proposed in Section 3. Our final method integrates dependency-aware structural constraints, which maintains stable throughput (≈ 150 FPS) across sparsity levels. This hybrid approach safely bypasses the slowdown region to maintain ~ 150 FPS on an NVIDIA RTX 4090 GPU even at extreme 50% sparsity compressions. Note that inference speed plateaus at ~ 150 FPS despite extreme backbone sparsity because the unpruned SSD multi-scale detection heads and CPU-bound Non-Maximum Suppression (NMS) become the dominating latency bottlenecks. These results indicate that accurate importance estimation alone is insufficient for deployable pruning; enforcing hardware-aligned structural constraints is equally critical.

4.5 Qualitative Interpretability

Qualitative visualization results using Spatial-Sensitive Grad-CAM are provided in the supplementary material, illustrating improved spatial consistency of retained feature maps when using our gradient-guided structural reduction.

4.6 Ablation Studies

We conduct controlled ablation experiments to isolate the contribution of each component in our framework: (i) the importance criterion, (ii) calibration subset size, (iii) structural constraints, and (iv) the role of fine-tuning. All experiments are performed on ResNet-50 trained on CIFAR-10 using identical pretrained checkpoints. Unless otherwise specified, results are reported as mean $\pm$ standard deviation over three random seeds $\{42, 123, 2024\}$.

Importance Criterion. We compare three channel importance metrics: L1-norm, absolute gradient-activation interaction ($|\nabla L \odot A|$), and the proposed squared gradient-activation formulation ($(\nabla L \odot A)^2$). All methods are evaluated under identical pruning ratios and followed by 5 epochs of fine-tuning to ensure stable comparison.

Table 1: Comparison of importance criteria on ResNet-50 (CIFAR-10). Results reported as mean $\pm$ std over 3 seeds.

Method	Sparsity	Accuracy (%)	FPS
L1-Norm	30%	95.55 $\pm$ 0.01	226.3 $\pm$ 4.2
Abs Grad-Act	30%	95.68 $\pm$ 0.09	231.3 $\pm$ 4.6
Ours (Squared)	30%	95.68 $\pm$ 0.14	229.1 $\pm$ 3.8
L1-Norm	50%	93.30 $\pm$ 0.14	233.9 $\pm$ 4.6
Abs Grad-Act	50%	94.26 $\pm$ 0.07	236.8 $\pm$ 2.6
Ours (Squared)	50%	94.24 $\pm$ 0.20	232.5 $\pm$ 2.3

At 30% sparsity, our method achieves $95.68 \pm 0.14\%$ accuracy, matching or slightly exceeding competing approaches. At 50% sparsity, our method achieves $94.24 \pm 0.20\%$, significantly outperforming L1-based pruning. While the squared formulation improves stability in most settings by reducing gradient sign cancellation, we observe that the absolute variant can perform competitively at extreme sparsity levels. This suggests that both formulations capture complementary aspects of channel importance, with the squared variant providing more consistent behavior across calibration samples.

Calibration Subset Size. We evaluate the sensitivity of our method to the size of the calibration subset using $\{10, 50, 100, 500\}$ samples at 30% sparsity. Performance remains stable across all settings, with accuracy varying within a narrow range ($95.52\% - 95.88\%$). Notably, even with as few as 10 samples, the model achieves $95.52 \pm 0.14\%$, demonstrating that a very small calibration subset is sufficient.

The runtime for importance estimation scales sub-linearly with calibration size, remaining below 1 second in all cases (e.g., ~ 0.25 s for 10 samples and ~ 0.5 s for 500 samples). This validates the efficiency of the proposed method and supports its use in fast post-training compression pipelines.

Structural Constraints. We evaluate the necessity of dependency-aware and group-aware pruning by comparing three variants: (i) naive channel pruning without constraints, (ii) dependency-aware

pruning (DepGraph), and (iii) our full method combining Grad-Act scoring with dependency and grouping constraints.

Naive pruning consistently produces invalid models due to channel mismatches in residual connections, resulting in runtime errors (e.g., incompatible tensor dimensions in convolutional layers). In contrast, both dependency-aware and full methods produce valid models with comparable computational cost. These results demonstrate that enforcing structural constraints is essential for maintaining architectural consistency and enabling deployable pruning. We note that this evaluation is performed prior to fine-tuning to isolate structural validity from optimization effects.

Impact of Fine-Tuning. We analyze the effect of post-pruning fine-tuning by comparing performance with and without 5 epochs of retraining. Without fine-tuning, all methods exhibit severe accuracy degradation ($\sim 10\text{--}15\%$), indicating that structural pruning introduces significant perturbations to the learned representations.

After fine-tuning, performance is largely recovered across all methods. At 30% sparsity, our method achieves 95.68%, comparable to L1 (95.51%) and DepGraph (95.87%). At 50% sparsity, our method achieves 94.24%, outperforming L1 (93.16%) and remaining competitive with DepGraph (94.68%). Importantly, the relative ranking between methods is preserved after fine-tuning, indicating that the proposed importance criterion leads to consistent pruning decisions.

5 Conclusion

We presented **Taylor-Made Sparsity**, a structured pruning framework that combines a data-driven, first-order Taylor proxy with hardware-aware dependency modeling to enable efficient and practically deployable model compression.

Our results demonstrate that while magnitude-based pruning often fails to preserve functional importance at high sparsity, gradient-based sensitivity estimation provides a more reliable signal for identifying critical channels. When integrated with dependency-aware structural constraints, this enables consistent reductions in computational cost while maintaining competitive accuracy across both classification (ResNet-50 [He et al., 2016] on CIFAR-10 [Krizhevsky, 2009]) and object detection (MobileNetV2-SSD [Sandler et al., 2018, Liu et al., 2016] on MS COCO 2017 [Lin et al., 2014]) tasks.

Importantly, our experiments show that accurate importance estimation alone is insufficient for deployable pruning. Without enforcing structural alignment, naive pruning leads to severe inference slowdowns due to hardware-level inefficiencies. By explicitly incorporating dependency modeling, our method ensures that theoretical sparsity translates into stable runtime performance.

Limitations. Our evaluation uses a short 5-epoch fine-tuning schedule to isolate pruning robustness rather than full convergence. While this enables controlled comparison across methods, longer retraining may further improve absolute performance, particularly for complex detection tasks. Furthermore, we emphasize that all inference throughput (FPS) measurements are conducted on an NVIDIA RTX 4090 GPU for controlled benchmarking. While the observed trends reflect structural efficiency improvements, absolute latency may differ on edge devices. Evaluating performance on representative edge hardware rather than desktop-class GPUs remains an important direction for future evaluation. Additionally, qualitative interpretability results using Spatial-Sensitive Grad-CAM [Selvaraju et al., 2017] are provided in the supplementary material rather than the main text due to space constraints.

Future Work. Extending this framework to dynamic sparsity, hardware-specific latency modeling, and evaluation on real edge devices (e.g., ARM-based systems or embedded accelerators) represents a promising direction for future research.

Broader Impacts. By compressing object detectors for edge devices, this work improves accessibility of real-time perception for low-power applications and reduces the computational and energy footprint of AI inference. However, efficient surveillance technologies inherently carry dual-use implications for privacy, necessitating responsible deployment frameworks.

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417 **Answer:** [Yes]

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419 Made Sparsity are clearly stated in the Introduction and supported by empirical results
420 in Section 4.

421 2. Limitations

422 **Question:** Does the paper discuss the limitations of the work performed by the authors?

423 **Answer:** [Yes]

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425 are discussed in Section 5.

426 3. Theory assumptions and proofs

427 **Question:** For each theoretical result, does the paper provide the full set of assumptions
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429 **Answer:** [Yes]

430 **Justification:** The Taylor approximation and assumptions are derived in Sections 3.1
431 and 3.2.

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433 **Question:** Does the paper provide open access to the data and code with sufficient
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